Information Loss at the CFAR Detection Stage for Indoor mmWave Human Activity Recognition

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Summary

Conventional CFAR is known to lose target information on extended targets, and recent work on indoor mmWave point clouds has argued that this loss, rather than aperture limits alone, is a major cause of point cloud sparsity. That claim is generally asserted rather than measured. This report measures it.

Using six synchronized radar and RGB-D indoor recordings, I sweep three detectors — 2D CA-CFAR, 2D OS-CFAR, and a Doppler-spread CFAR after Zhang — across nominal pfa, and compare the resulting point clouds against a fixed-power-threshold oracle that marks the reference level of recoverable target energy. The result is that the information is recoverable but expensive: OS-CFAR does reach the oracle target count, and only at a false alarm load that makes the point cloud unusable/noisy downstream. The loss is therefore not a hard limit of the detector, it is a price set by the fixed two-dimensional reference window geometry acting on a cell under test belonging to a target whose range extent is predictable and whose Doppler extent is not.

1. Motivation

The point cloud is the primary data representation for mmWave radar sensing, and human activity recognition is no exception. mmWave point clouds are sparse and low resolution, and the field has largely attributed this to aperture limits. That attribution has pushed research toward data-driven methods that learn around the sparsity under supervision from a higher resolution sensor — point cloud super resolution and human mesh reconstruction [2–7].

My position is that the processing pipeline's own contribution to this sparsity is overlooked. Beyond the aperture limit, the pipeline is responsible for condensing dense I/Q samples into a point cloud representation, and each stage of that condensation loses information. Out of these stages, RD map adaptive detection (CFAR) is prone to undetected bins, ghost detections, and frame-to-frame inconsistency. This is the stage this report investigates.

The work started from ETCM-CFAR [1], which argues that prior work underestimates CFAR's influence on point cloud quality, and that spatial CFAR suffers a severe energy masking problem indoors because the human body at short range is an extended target rather than a point. RadarHD [2] arrives at a related position from the other direction: it deliberately selects a low CFAR threshold so as not to discard target information, and accepts the resulting ghost points as the cost.

Both treat the loss as established. Neither measures how much information is actually lost, or what it costs to get it back. That is the gap this report addresses.

The thesis: the CFAR stage is an under-examined information bottleneck for indoor HAR. CFAR designs are universal and general purpose by intention. A human indoors is a very specific scatterer in a very specific multipath clutter environment, and the mismatch between the two costs target information — the low-SNR Doppler skirt goes undetected — and injects instability in the form of ghosts and frame-to-frame inconsistency.

2. Background: how the field has handled CFAR information loss

CFAR information loss on extended targets is well established and long predates indoor mmWave. The standard attribution is reference window contamination: energy from the extended target itself leaks into the training cells, the local background estimate is inflated, the threshold rises, and the target masks itself. The problem has been documented and attacked from naval radar through automotive radar.

The solutions fall into two families:

- **Geometry.** Resize the guard and reference cell structure so that the extended target sits inside the guarded region rather than inside the training region.
- **Estimator.** Replace cell averaging with a statistic that is less sensitive to a few contaminated reference cells — order statistic, greatest-of, smallest-of, trimmed mean, and the rest of the family.

Both families keep the closed-form threshold and, with it, the assumption that the reference cell background follows an exponential PDF. Of these, OS-CFAR is the most consistently robust across multitarget and dynamic scenes, and it is the practical baseline for this work.

The obvious response to Doppler-axis masking is to enlarge the Doppler reference window to accommodate the large human doppler spread that is created in dynamic cases. This carries three costs. First, compute: OS-CFAR sorts the reference set per cell under test, so cost scales with window size. Second, reduced locality, which is the entire point of an adaptive threshold. Third, a larger window has a higher probability of being contaminated by other targets and by multipath. And shown by the authors of ETCM CFAR, it does not fully solve the problem in any case — OS-CFAR still misses the low-SNR Doppler skirt.

This gives three distinct failure modes, which are the object of this report:

1. Target bins undetected because the extended target + indoor multipath makes the CUT reference window non-homogeneous.
2. The low-SNR Doppler skirt is specifically undetected, masked relative to the high-SNR torso bulk and surrounding multipath.
3. Nominal PFA is not realized PFA. The closed-form threshold delivers the designed false alarm probability only if the reference cells actually follow the assumed exponential PDF. Indoor clutter does not, so the designed operating point is not the operating point.

Failure modes 1 and 2 are measured in this report. Failure mode 3 is theoretically sound but is not quantified here; Section 9 states why.

3. The human as a scatterer

A human is an extended target and a complex scatterer with a diverse Doppler profile. From visual inspection of RD maps across the indoor datasets collected for this work, together with Rohling's characterization of pedestrians [8], the structure that matters for detection is:

- **Torso.** High-SNR bulk Doppler. This is the reliable hotspot that any reasonable threshold catches.
- **Limbs and skirt.** A low-SNR, Doppler-diverse smear around the bulk. This is where the HAR-relevant information lives and it is what goes undetected under conventional CFAR. It is genuine target energy, not noise: when these points are detected, they project into the target region of interest against depth ground truth. The mechanism is reference window masking relative to the torso.
- **Spread asymmetry.** The Doppler spread is large and varies frame to frame with gait phase, pose, and limb activity. The range spread is comparatively predictable — a frontal human occupies roughly 0.3–0.5 m in range, dominated by the torso, which is the dominant contributor to target RCS.

That asymmetry is the premise for everything in Section 5.3. Range extent can be designed for. Doppler extent cannot.

I do not attempt to attribute individual RD bins to torso versus limbs, because the radar phase is stochastic [9] and the pipeline carries no bin-to-point bookkeeping. For this work the evaluation is simply target or not target.

4. The indoor environment

4.1 Static clutter and MTI

For static indoor clutter, MTI removal is the best return on investment. It eliminates non-homogeneous static clutter in one cheap step. More sophisticated alternatives such as PCA-based subspace removal exist but are inconsistent in practice because their performance depends on the quality of the constructed covariance matrix [was tested and did not perform consistently].

MTI comes at a real cost. Quasi-static humans drop significantly in SNR after mean subtraction, which makes them prone to masking and can leave them entirely undetected at low PFA. This is the same failure mechanism as the skirt — real target energy sitting below the local threshold — reached by a different route. Because of this, both MTI states are reported everywhere in this report rather than fixing one. Figure below shows a frame selected that shows undetected points for a static person at 2m due to lower SNR for the doppler spread CFAR and OS CFAR methods.

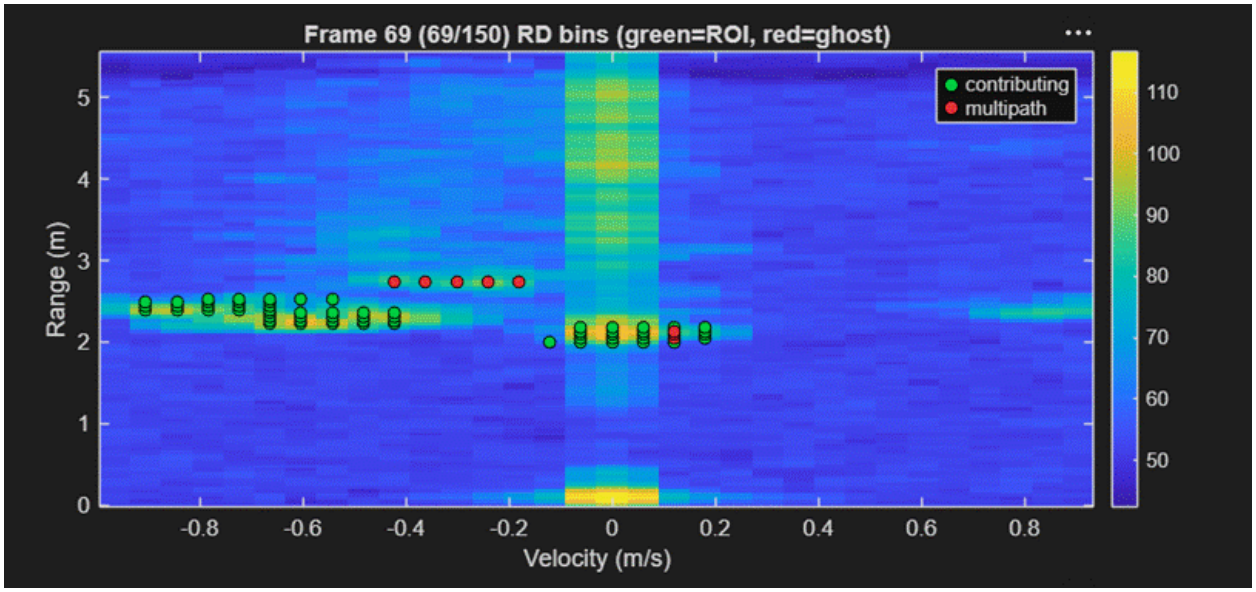


Fig 2.a: Doppler Spread CFAR Static removal off. Higher SNR at 0 doppler, hence more detections

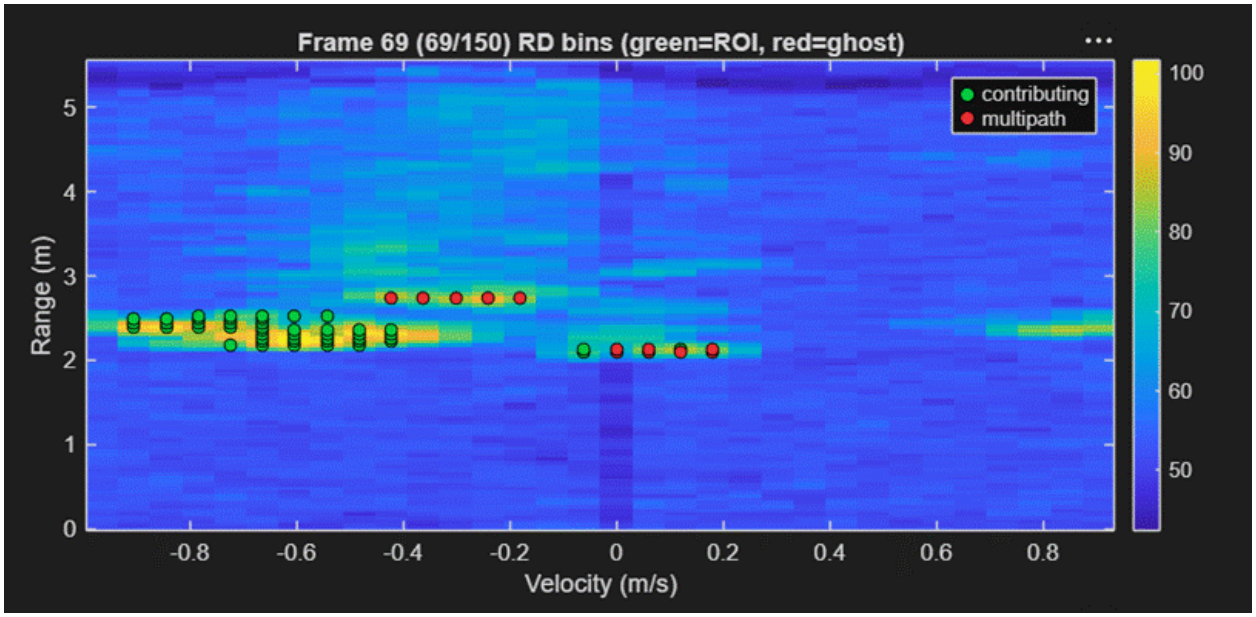


Fig 2.b: Doppler Spread CFAR Static removal on. Less SNR at 0 doppler due to MTI; mis detected and less detected points.

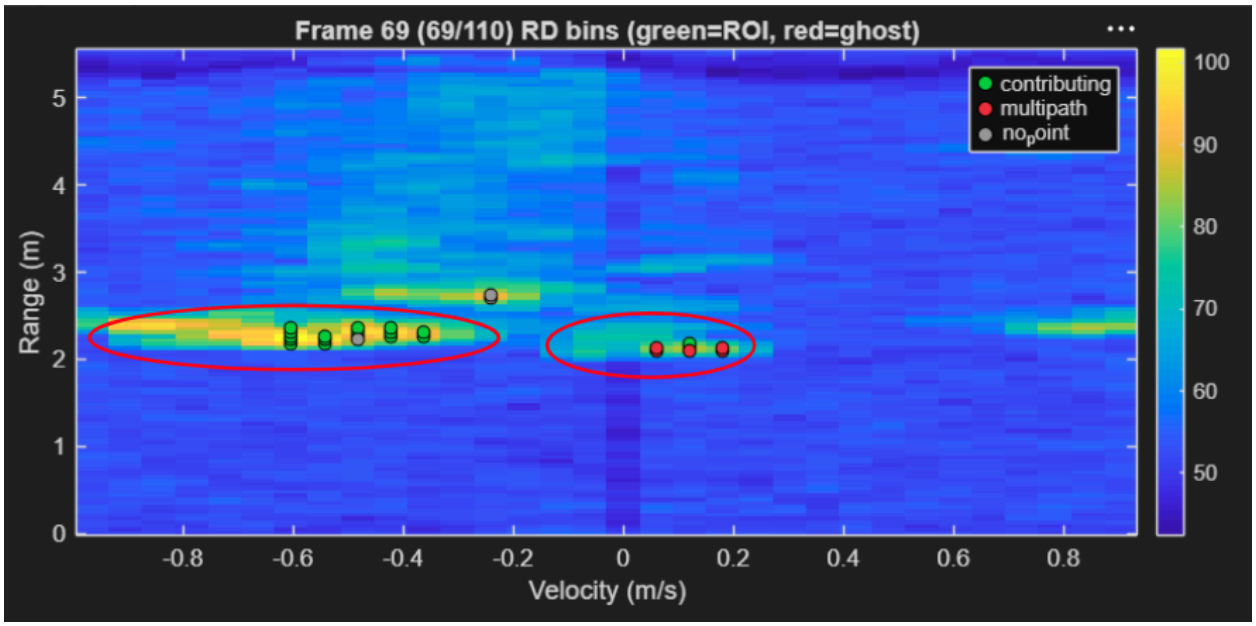


Fig 2.c: OS CFAR Static removal on. Less SNR at 0 doppler due to MTI; mis detected and less detected points.

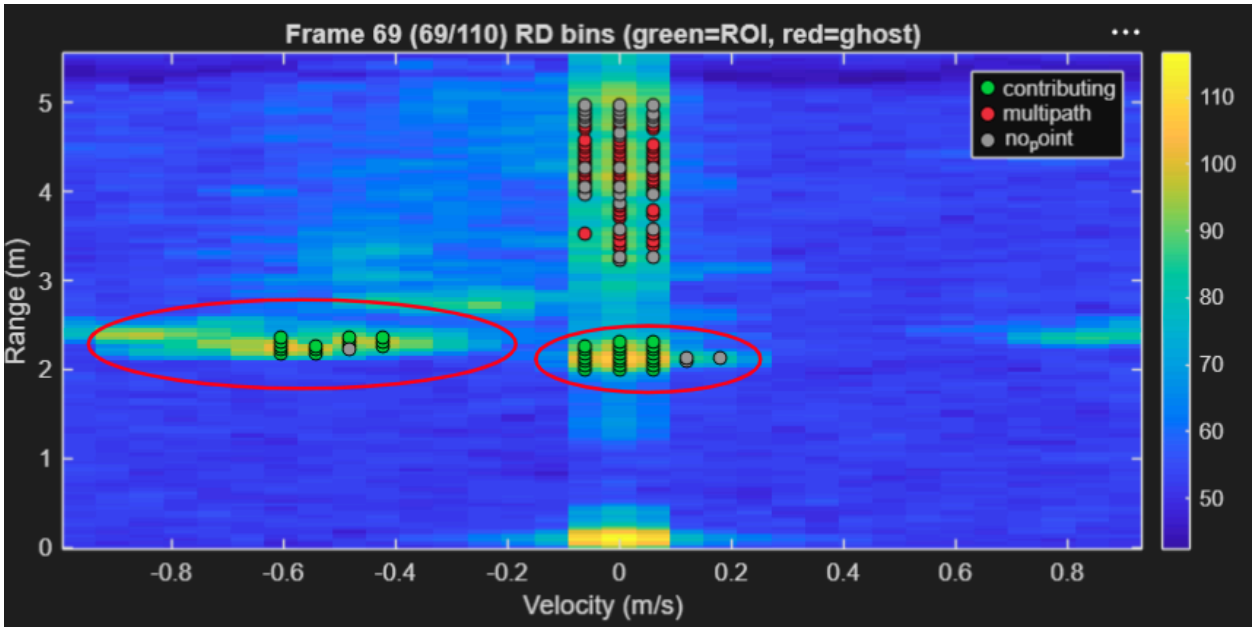


Fig 2.d: OS CFAR Static removal off. Higher SNR at 0 doppler, hence more detections

4.2 Dynamic, target-coupled multipath

The distinctive challenge for indoor HAR is that multipath is a function of the moving human target and of the specific room. Multipath returns inherit Doppler smears similar to the target's but appear at extended ranges, and indoors they can carry significant SNR. The consequence for detection is that the probability of a reference cell being contaminated rises, which lowers the realized PFA relative to design.

A terminology point worth making explicit, because it is a common source of confusion: multitarget returns and multipath are clutter by any ordinary definition, but they are not the clutter that appears in the CFAR mathematical model. The exponential assumption models a constant-PDF noise background whose probability decays exponentially with power. It exists because it is what makes a closed-form threshold possible. Target-coupled multipath is neither constant nor exponential — it is bursty, scene dependent, and correlated with the target. This violation of the assumed exponential PDF does not guarantee the designed PFA; as developed in Section 9, it in fact pushes realized false-alarm behaviour in two opposing directions depending on whether the heavy tail is viewed from the reference cells or from the cell under test.

Work that examines indoor scattering profiles and the failure of the exponential assumption under multipath [9, 10] converges on the same conclusion: indoor multipath cannot be modeled by a single homogeneous PDF. Safa et al. [10] respond by dropping the assumption entirely in favour of a low-complexity PDF-free detector; Chizhik et al. [9] provide backscatter measurements in cluttered environments that support the heavy-tailed characterization. This is the crack in the foundation of the CFAR design that Section 5 builds on.

5. CFAR as a design skeleton, and where the human breaks it

5.1 Anatomy

Every CFAR reduces to a threshold $T = \alpha \cdot Z$ applied to a cell under test, with fixed guard cells and fixed reference cells around it. Clutter assumptions enter in exactly two places:

- **The scale factor α** is a function of the desired PFA, the assumed clutter PDF, and the statistic used for Z. The exponential assumption is what makes α a clean closed form, and it is exactly what indoor HAR violates.
- **The local clutter estimate Z** is a statistic over the reference cells. This is where most of the literature lives — CA, OS, GO, SO and their variants, each with different behaviour at clutter edges, on extended targets, and under reference cell contamination, much of it driven by automotive radar research. Nearly all of it retains the exponential PDF and works to make Z track that assumption more closely. OS CFAR with its kth order statistic manages to maintain reasonable performance under this assumption.

5.2 Exploration of prior work, and where my two research dimensions sit

I organize prior CFAR work into three branches. This taxonomy is the skeleton for deciding what to read, what to test, and what to build.

- **(a) Estimator variants under the exponential assumption.** CA, OS, GO, SO and family. Vary Z, keep the PDF. My CA versus OS comparison lives here and reconfirms the known result that OS is more robust in multitarget and dynamic scenes.
- **(b) Test statistic restructuring under the same assumption.** Zhang et al. [14] is the key reference. Their detection statistic is a noncoherent sum of D Doppler bins per range cell fed to a 1D range

OS-CFAR. Zhang modifies the test statistic and its geometry around the Doppler-spread target, but not the clutter model. Wei et al. [11] address the single-cell CUT failure mode with an area-based CFAR for extended targets; the useful takeaway is that a larger cell under test encapsulating the extended target detects better because of accumulated SNR, though it uses coupled range–Doppler cells at a fixed group size. This is **dimension 1 of my research: restructuring the detection geometry around the human scatterer, mainly in the direction proposed by Zhang.**

- **(c) PDF-free and robust methods.** Safa et al. [10] drop the exponential assumption to gain robustness. Levanon and Shor [12] examine OS-CFAR under a Weibull background and conclude it works on heavy-tailed PDFs provided the shape parameter is known or estimated, recommending the Weber and Haykin approach [13] where the shape varies substantially. This is **dimension 2 of my research: questioning the clutter PDF itself for indoor multipath and multisubject cases.**

5.3 Why the 2D window geometry cannot be tuned universally for HAR

On the Doppler axis, the spread produced by a moving human has high variance across frames — this follows from Rohling's pedestrian characterization and from my own RD maps. No single fixed window size satisfies it. Sizing for the widest case sacrifices locality and invites contamination; sizing for the typical case leads to undetected bins due to contamination.

The range axis is different. At 0.044 m per bin, a 0.3–0.5 m human spans roughly 7–11 range bins. My guard span matches that target extent and my training window sits plausibly outside the body. Range geometry can be designed once and will hold.

This asymmetry is the origin of the decoupling idea: remove the Doppler domain from the CFAR entirely, condense it per range bin, and run threshold-based detection only on the predictable range profile. The probability that extended-target energy contaminates the reference cells drops, because the dimension where that contamination is unpredictable is no longer part of the background estimate. Zhang's method demonstrates this benefit indirectly, though their motivation was long-range SNR integration rather than indoor clutter.

5.4 The Zhang method as implemented

Zhang's statistic: for each range cell, sum the energy of D contiguous Doppler bins, slide that window over Doppler positions, and take the maximum of the summed energy. The resulting per-range statistic goes to a 1D OS-CFAR over range, with α derived from the chi-squared 2D-DOF distribution.

I implemented the sliding-window variant using the same range guard and training geometry as my OS-CFARconfiguration, so that the comparison isolates the Doppler treatment. One deliberate deviation made from the paper: their D = 10 at 0.0122 m/s per bin covers roughly 0.12 m/s of Doppler. My resolution is 0.07 m/s per bin, so a physically equivalent window would be $D \approx 2$, which is meaningless against a human skirt spanning most of my Doppler axis. Baseline runs therefore use D = 5, sized by an estimate based on average doppler spread of humans.

Section 8.3 shows that D = 5 was undersized, and that the window sizing rather than the decoupling itself is what limits the method. This is the central interpretive point of the results and is developed there.

6. Experimental setup

6.1 System

A TI AWR6843ISK Rev. D with a DCA1000EVM capture card, mounted next to an Intel RealSense D455 depth camera. Radar frames were captured at 10 frames per second; depth at approximately 30 frames per second at 640 × 480. The two sensors are aligned in coordinates by a physically measured x-shift, and in time by a physical trigger — a foil-covered box pushed toward the radar, with the resulting Doppler energy spike aligned by inspection against the visual depth stream.

6.2 Chirp configuration

The chirp was designed for high range resolution while balancing Doppler resolution against maximum unambiguous velocity for basic indoor human activity. Given the tradeoffs involved, that balance was reached with a compromised maximum Doppler.

Parameter	Value
Range resolution	0.044 m
Maximum unambiguous range	9.02 m
Maximum radial velocity	±1 m/s
Radial velocity resolution	0.07 m/s
Frame duration	100 ms
RD map size	128 range bins × 32 Doppler bins

6.3 Datasets

Six synchronized radar and RGB-D recordings, each 15 seconds and 150 radar frames.

ID	Scene	Frames evaluated	Time offset	Notes
c	Two people: static subject at 2 m on boresight; second subject walking 3 m → 1 m at approximately 30°	111 (40–150)	+1.6 s	Mixed static–dynamic. Known possible field-of-view bias.
d	Two people: subject walking 3 m → 1 m on boresight; static subject at 2 m and approximately 30°	106 (45–150)	+2.0 s	Mixed static–dynamic, boresight and off-axis roles reversed relative to c. Used for the Doppler-width sweep.

e	Two-person reciprocal range motion: one subject 3 m → 1 m, the other 1 m → 3 m on the opposite side	106 (45–150)	+2.0 s	Dynamic multi-person. Richest multipath case.
f	Two-person reciprocal range motion, trajectories reversed relative to e	106 (45–150)	+2.0 s	Dynamic multi-person.
g	One static subject in a starfish pose at 2 m	106 (45–150)	0 s	Static single person. Entire segmented subject cloud used as ground truth.
h	One person walking 1 m → 3 m	106 (45–150)	+2.1 s	Dynamic single person.

In total: 900 recorded radar frames, 641 radar frames evaluated per detector configuration, and 3,221 RealSense frames.

6.4 Settings held constant

- Radar mounting correction: range_roll_180.
- Camera-to-radar translation: +0.1143 m along x.
- Depth-person radial range gate: 0–4 m.
- Radar/depth matching: nearest host timestamp after applying the dataset-specific time offset.

7. Evaluation methodology

The question is how many target points each detector recovers, and how many ghost points it has to admit in order to recover them. The cost of target information is therefore expressed as ghost points per target point. The preprocessing pipeline intentionally involves a post AoA DBSCAN stage that is mainly added to prove that the ghost points measured cannot be removed by a conventional DBSCAN stage that is used extensively in mmWave radar point cloud research. The cost of target information is read from a PFA sweep operating curve. Everything downstream of the detector is held fixed, so differences between curves come from the detector, the MTI state, the PFA, or the Doppler window width.

7.1 Radar preprocessing

- Blackman window and 128-point FFT along range.
- Two clutter conditions evaluated throughout: MTI off (no static clutter subtraction) and MTI on (mean static clutter subtraction across chirps, TDM-MIMO structure preserved).
- Hann window and 32-point FFT along Doppler.
- Detector-selected RD bins passed to the same angle estimation and post-processing.

7.2 Detectors compared

Three detectors: two-dimensional CA-CFAR, two-dimensional OS-CFAR, and the Doppler-spread OS-CFAR after Zhang (DS-CFAR) at D = 5. Each is evaluated with MTI on and off across ten logarithmically spaced nominal PFA values from 1×10^{-6} to 0.3.

Parameter	Value
Range guard bins	4
Doppler guard bins (CA/OS)	2
Range training bins	8
Doppler training bins (CA/OS)	4
OS rank fraction	0.75
Edge policy	Skip incomplete edge windows
Peak grouping	Disabled

7.3 Doppler-width experiment

To test whether DS-CFAR's target recovery is limited by the fixed Doppler integration width, dataset d is evaluated at $D \in \{3, 5, 10\}$ over the full PFA grid and both MTI states, so D is the only detector parameter varied. The Doppler integration uses a linear sliding window; windows do not wrap between the last and first Doppler bins.

The hypothesis is judged by target recovery at comparable realized ghost rates. If a larger D yields more target points without a matching rise in ghosts, that supports the reading that the integration width was the binding constraint. Improvement at the same nominal PFA alone would be weaker evidence, since realized false alarm behaviour varies with D.

7.4 Oracle reference

The oracle is a fixed absolute power threshold of 70 dB received power applied directly to the RD map. Every bin above it is retained and passed through the identical angle estimation, DBSCAN filtering, and depth-truth scoring as the detectors. It is not a deployable detector — it is a reference for how much useful target information still sits below the thresholds the CFAR detectors pick.

The value was chosen after inspecting the data so that it sits at the power level where the Doppler skirt lives: high enough that noise floor bins do not inflate the target point count, low enough that the skirt is included. It is applied identically to every frame and every dataset, and appears as a single marker on each operating curve rather than as a sweep. This still does not exactly capture the skirt for all datasets but is on the conservative side to prevent the noise floor from being converted to points.

An earlier version of this evaluation defined the oracle relative to the per-frame peak power, retaining all bins within n dB of the peak. That was dropped in favour of a fixed absolute threshold because the frame-relative rule produces inconsistent ghost behaviour. The retained set moves with whatever the peak happens to be in a given frame: a frame with a closer strong torso return raises the effective floor and discards skirt bins, while a weak far away target lowers it and admits noise. Ghost counts under that

rule are therefore not comparable across frames or datasets, which defeats the purpose of a reference marker sitting on an operating curve whose x-axis is a ghost count. A fixed absolute threshold gives a stable reference and makes the oracle directly comparable to the detector curves it is plotted against.

7.5 Angle estimation and point cloud filtering

Identical for every detector: OmniPC/MUSIC angle estimation, maximum two targets per RD bin, 8 dB angle-peak threshold, Cartesian crop disabled, post-AoA DBSCAN with radius 0.30 m and minimum 4 points. Because this stage is shared, differences in the operating curves come from the detector, the MTI state, the PFA, or D. Absolute point counts carry AoA noise; relative rankings do not.

7.6 Depth-based ground truth

For each radar frame the nearest synchronized RealSense depth frame is selected after applying the manual time offset. The person depth cloud is restricted to 0–4 m radially, cleaned with a minimum connected-component size of 50 pixels, voxel-downsampled at 0.03 m, and transformed with the +0.1143 m x translation and the mounting correction.

A radar point is a target if its nearest depth-cloud point satisfies all three spherical tolerances below; otherwise it is a ghost. The tolerances account for time synchronization mismatch, coordinate alignment error, and cross-sensor translation error.

Measurement axis	Tolerance
Range	±0.20 m
Azimuth	±10°
Elevation	±5°

7.7 Metrics

For each frame, T is the number of post-DBSCAN radar points classified as target and G the number classified as ghost. Operating curves plot mean ghost points per frame on the x-axis against median target points per frame on the y-axis, with a shaded band showing the 25th–75th percentile of T. Solid lines are MTI on, dashed lines MTI off.

The asymmetry between mean and median is deliberate. Ghost points are clutter that must be eliminated downstream and their cost accumulates across frames, so the long-run average is the relevant quantity; the mean ghost count is also the direct analogue of the false alarm rate the PFA sweep controls. Target points are consumed per frame — pose, mesh, and clustering all require sufficient points within a single frame — so the relevant quantity is the yield of a typical frame rather than an average an outlier frame can contaminate. Comparisons are made by curve dominance, which is an ordering and is insensitive to the specific estimator.

8. Results

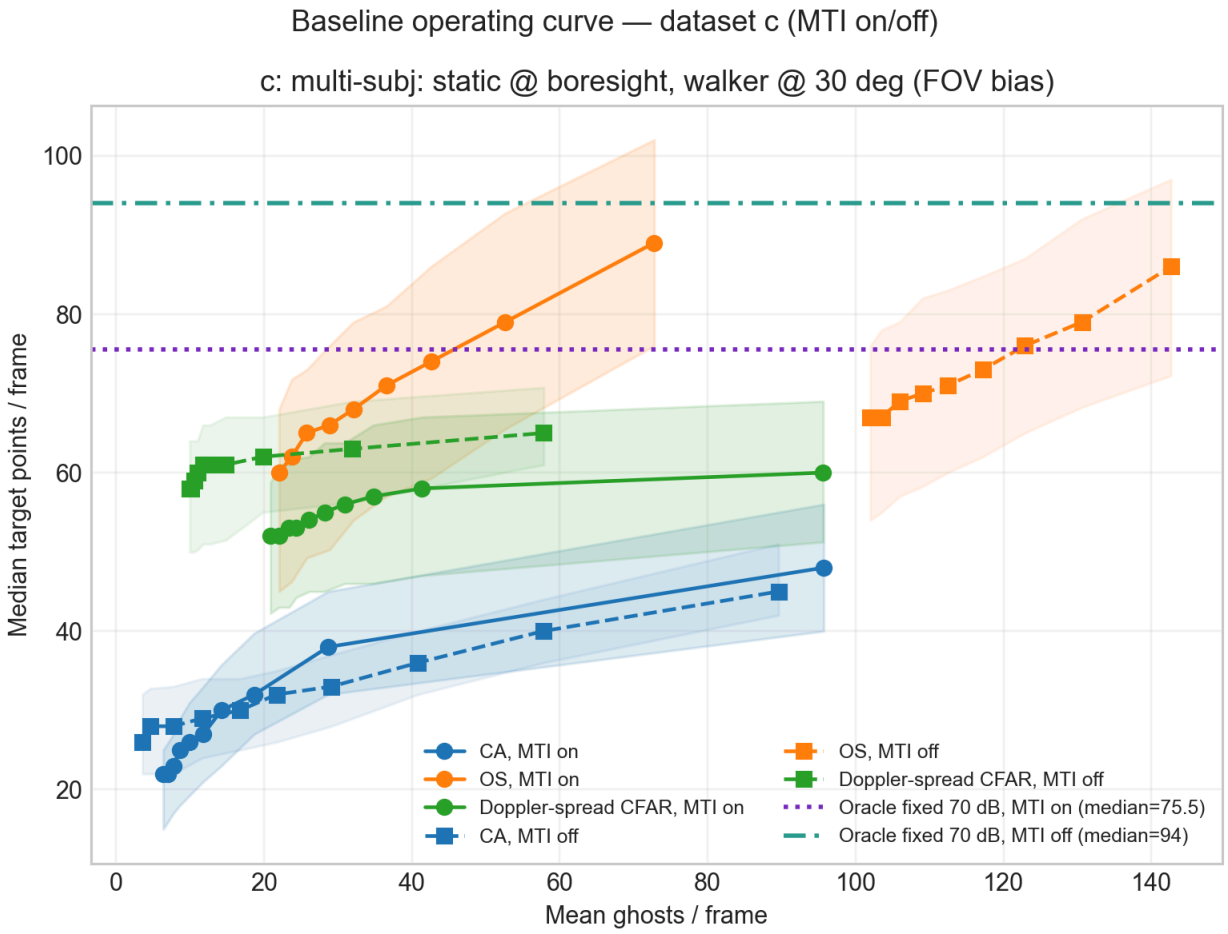
Reading the curves is straightforward: for a given detector, sweeping the PFA traces out how many target points it recovers against how many ghosts it admits. The detector whose curve sits higher and further left is better on that scene. The oracle appears as a single marker giving the recoverable reference level.

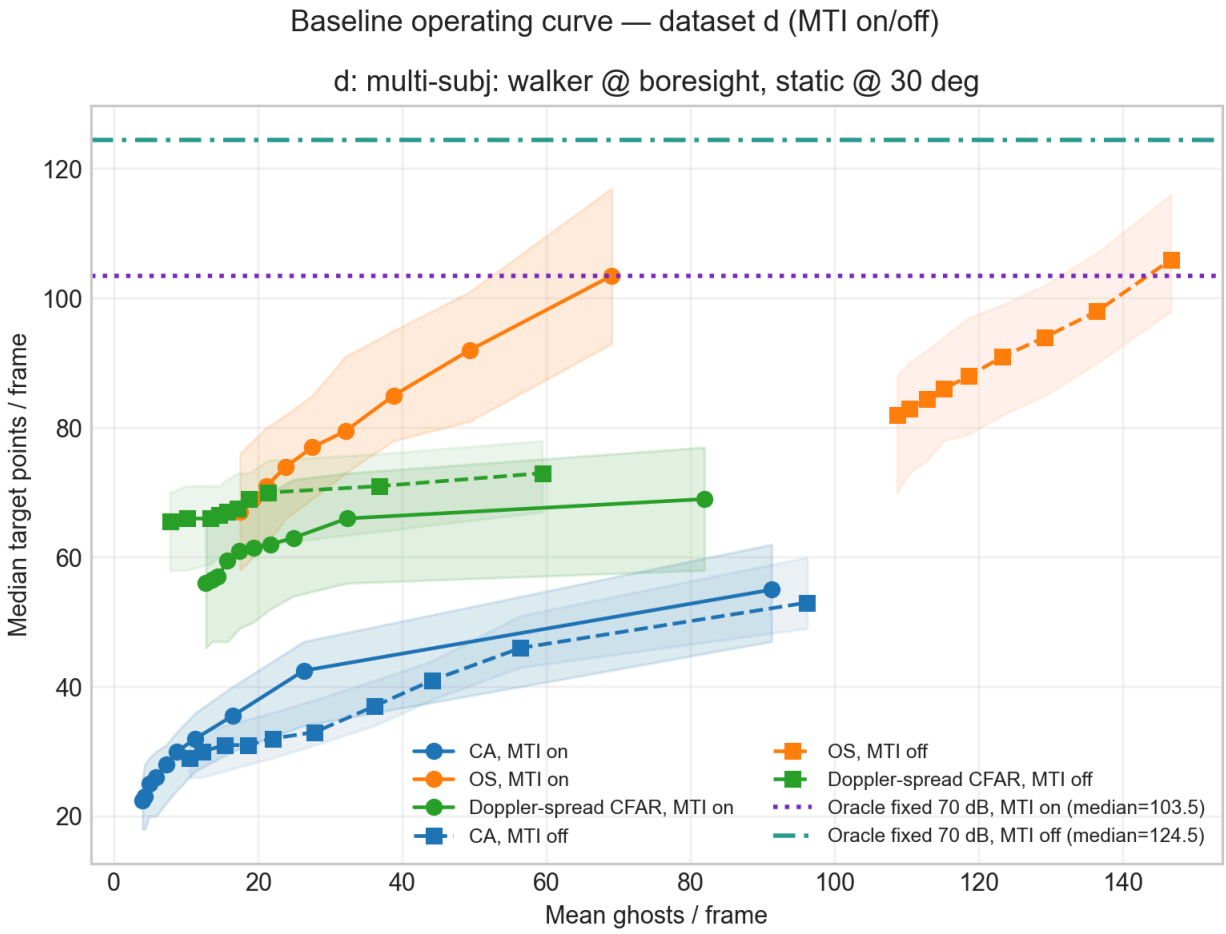
8.1 The oracle gap, and what it costs to close it

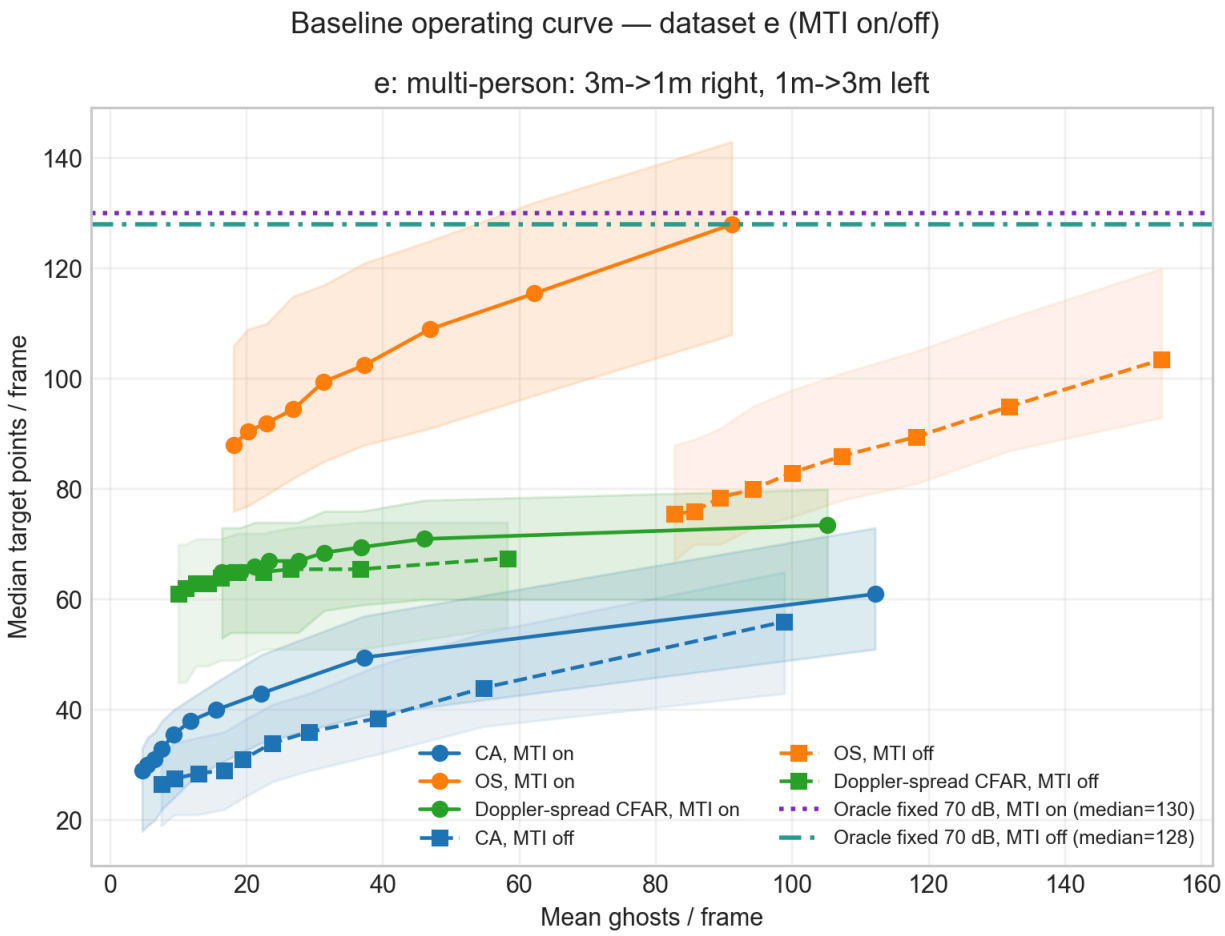
With this result in the plots it can seem that OS CFAR matches the oracle well, but the ghost points accepted by it are post DBSCAN hence cannot really be filtered. Additionally the MTI argument discussed in Section 8.5 exposes another potential downside of OS CFAR for indoor HAR. The effect of MTI on information loss in datasets containing quasi static targets can clearly be seen. Lastly, the OS CFAR compute per CUT is unreasonable for matching the 10 fps baseline for standalone radars without compromising the time interval between chirps, dropping the velocity resolution.

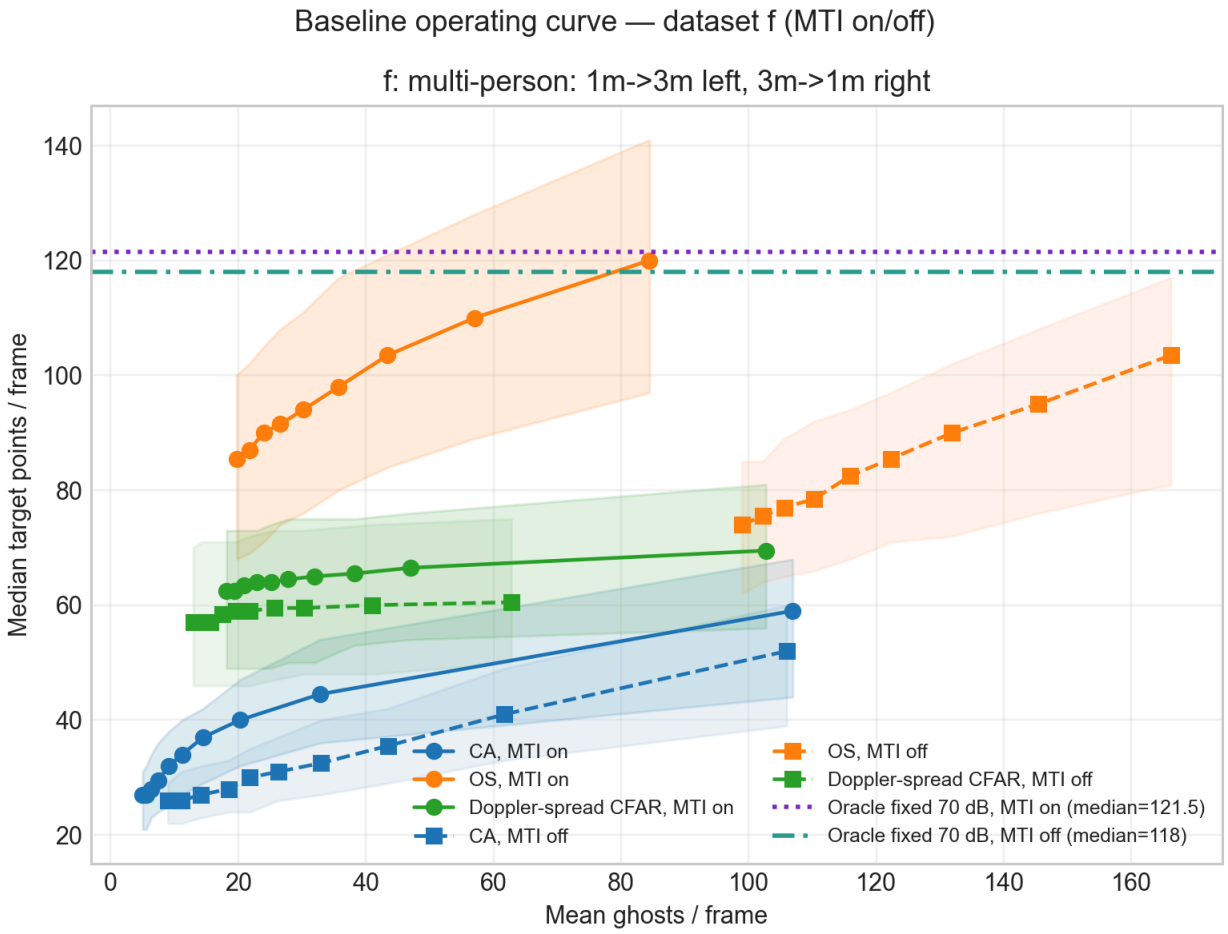
OS-CFAR does eventually reach the oracle target count, at the top of the PFA sweep, and it pays for it in significant ghost points even after DBSCAN filtering. The oracle is with a 70 dB threshold, although not exact to capture the skirt for all datasets, we can see the approximate target points per dataset and compare performance.

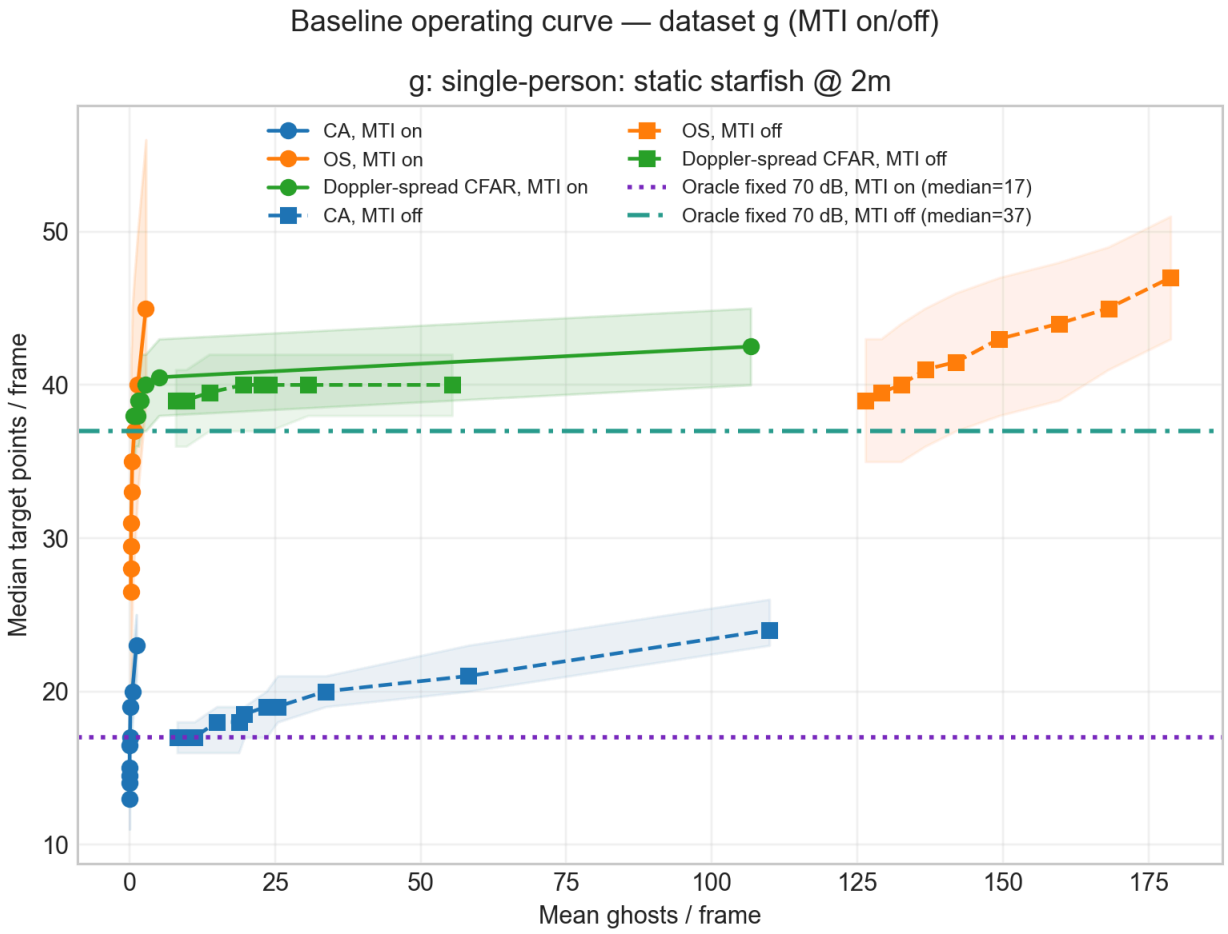
The skirt information is not beyond the reach of a conventional detector; it is behind a price. Conventional CFAR recovers it only by opening the threshold far enough that the resulting point cloud is dominated by false alarms, which is not a usable operating point for anything downstream. The loss is therefore a cost curve, not a hard ceiling, and the object of any improved detector is to move that curve rather than to discover information that was not there. Doppler spread CFAR moves the curve up when the fixed window is sized right, which shows potential in the method with drawbacks related to window sizing (Section 8.3 explores this further).


Baseline operating curve — dataset c (MTI on/off)
c: multi-subj: static @ boresight, walker @ 30 deg (FOV bias)
Median target points / frame
Mean ghosts / frame
100
80
60
40
20
0
20
40
60
80
100
120
140
CA, MTI on
OS, MTI on
Doppler-spread CFAR, MTI on
CA, MTI off
OS, MTI off
Doppler-spread CFAR, MTI off
Oracle fixed 70 dB, MTI on (median=75.5)
Oracle fixed 70 dB, MTI off (median=94)


Baseline operating curve — dataset d (MTI on/off)
d: multi-subj: walker @ boresight, static @ 30 deg
Median target points / frame
Mean ghosts / frame
0
20
40
60
80
100
120
140
20
40
60
80
100
120
CA, MTI on
OS, MTI on
Doppler-spread CFAR, MTI on
CA, MTI off
OS, MTI off
Doppler-spread CFAR, MTI off
Oracle fixed 70 dB, MTI on (median=103.5)
Oracle fixed 70 dB, MTI off (median=124.5)


Baseline operating curve — dataset e (MTI on/off)
e: multi-person: 3m->1m right, 1m->3m left
Median target points / frame
Mean ghosts / frame
0
20
40
60
80
100
120
140
160
20
40
60
80
100
120
140
CA, MTI on
OS, MTI on
Doppler-spread CFAR, MTI on
CA, MTI off
OS, MTI off
Doppler-spread CFAR, MTI off
Oracle fixed 70 dB, MTI on (median=130)
Oracle fixed 70 dB, MTI off (median=128)


Baseline operating curve — dataset f (MTI on/off)
f: multi-person: 1m->3m left, 3m->1m right
Median target points / frame
Mean ghosts / frame
140
120
100
80
60
40
20
0
20
40
60
80
100
120
140
160
CA, MTI on
OS, MTI on
Doppler-spread CFAR, MTI on
CA, MTI off
OS, MTI off
Doppler-spread CFAR, MTI off
Oracle fixed 70 dB, MTI on (median=121.5)
Oracle fixed 70 dB, MTI off (median=118)


Baseline operating curve — dataset g (MTI on/off)
g: single-person: static starfish @ 2m
CA, MTI on
OS, MTI on
Doppler-spread CFAR, MTI on
CA, MTI off
OS, MTI off
Doppler-spread CFAR, MTI off
Oracle fixed 70 dB, MTI on (median=17)
Oracle fixed 70 dB, MTI off (median=37)
Median target points / frame
Mean ghosts / frame
10
20
30
40
50
0
25
50
75
100
125
150
175

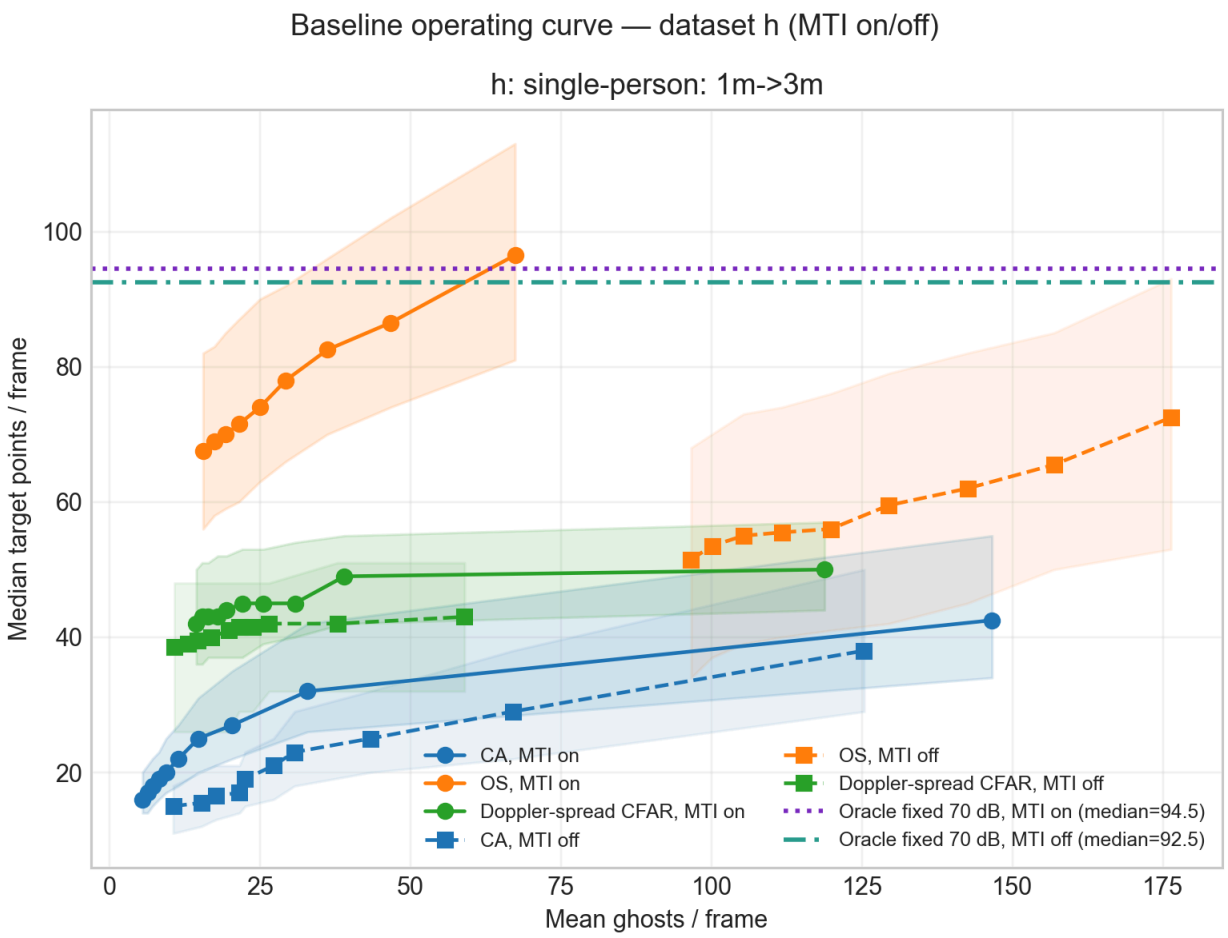


Fig. 3.1–6: Baseline operating curves, datasets c–h. Median target points versus mean ghost points across the PFA sweep, with the oracle marker. Solid = MTI on, dashed = MTI off.

8.2 Detector comparison

CA-CFAR is clearly the worst of the three across the dynamic and mixed scenes, as expected in multitarget conditions. OS-CFAR keeps recovering more target points as the threshold loosens.

DS-CFAR at D = 5 sits below OS-CFAR on every dataset. Its target count moves little as PFA rises while its ghost count climbs, producing a flat curve with a rising tail. 8.3 justifies DS CFAR's performance dependence on window sizing with evidence.

The mechanism is a loss of selection dimensionality. OS-CFAR is a genuinely two-dimensional detector: it exercises threshold control over both the range and the Doppler axis for every cell under test. DS-CFAR gives that up. It detects only along range bins, and the Doppler axis is collapsed by a fixed-window maximum, so for a given range bin the number of Doppler detections it can produce is capped by the window size. As PFA rises it cannot keep pulling in more target energy along Doppler; the additional detections are ghost points from range cells selected due to the higher PFA that hold no target. This behaviour can be taken advantage of if executed right, because it maintains the closed form pfa solution of CFAR along with lesser compute, and robustness to human doppler spreads.

8.3 Doppler-width sweep, and what it means for 8.2

Sweeping D ∈ {3, 5, 10} on dataset (d) over the same PFA grid, target point recovery increases significantly with the window width (Fig. 4.1).

This result is what makes Section 8.2 interpretable, and it should be read carefully. The underperformance of DS-CFAR in the baseline is not a verdict on decoupling range from Doppler. It is a verdict on the fixed window: D = 5 was undersized for the Doppler extent of a human at my resolution, and that alone caps target recovery. Widening the window recovers the points the narrow window left behind but is prone to accepting multipath, which means the integration width, not the range detector and not the decoupling, was the binding constraint. Correctly sized, a decoupled detector can exceed OS-CFAR.

It is worth being explicit about analysis here, because the baseline operating curve comparison invites the wrong reading. Implementing Zhang was never an attempt to show that an off-the-shelf Doppler-spread CFAR beats OS-CFAR. It was a test of whether the human Doppler spread needs to be handled as a dimension in its own right rather than as another axis of a 2D window. The evidence says it does, and that the fixed window is precisely what stops the existing instance of that idea from doing it well. Comparing the D = 10 curve in Fig. 4.1 against the OS-CFAR curve in Fig. 4.2, at approximately 100 target points the correctly sized Doppler window reaches that yield with roughly ~50% fewer ghost points than OS-CFAR.

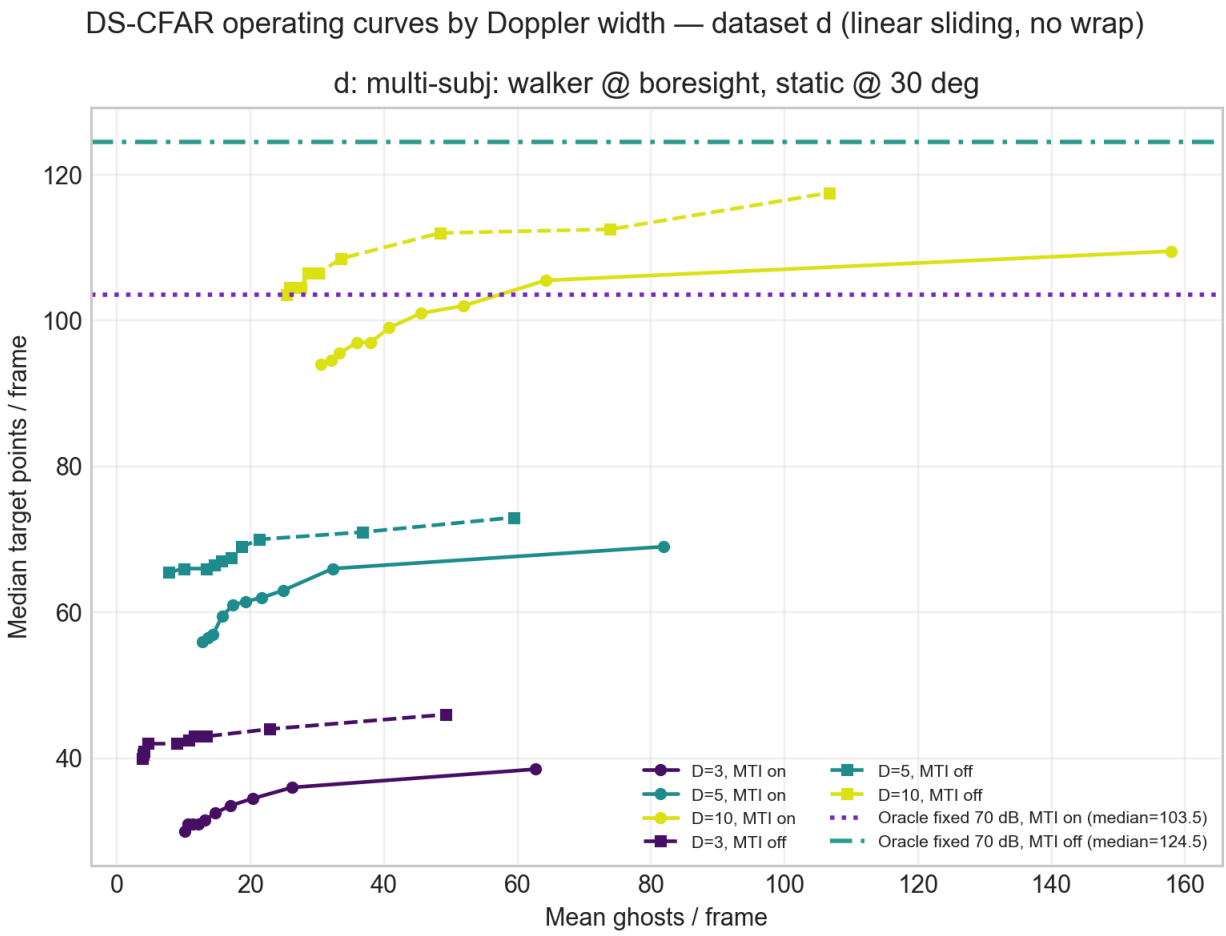


Fig. 4.1: DS-CFAR operating curves by Doppler width, dataset d (linear sliding window, no wrap).

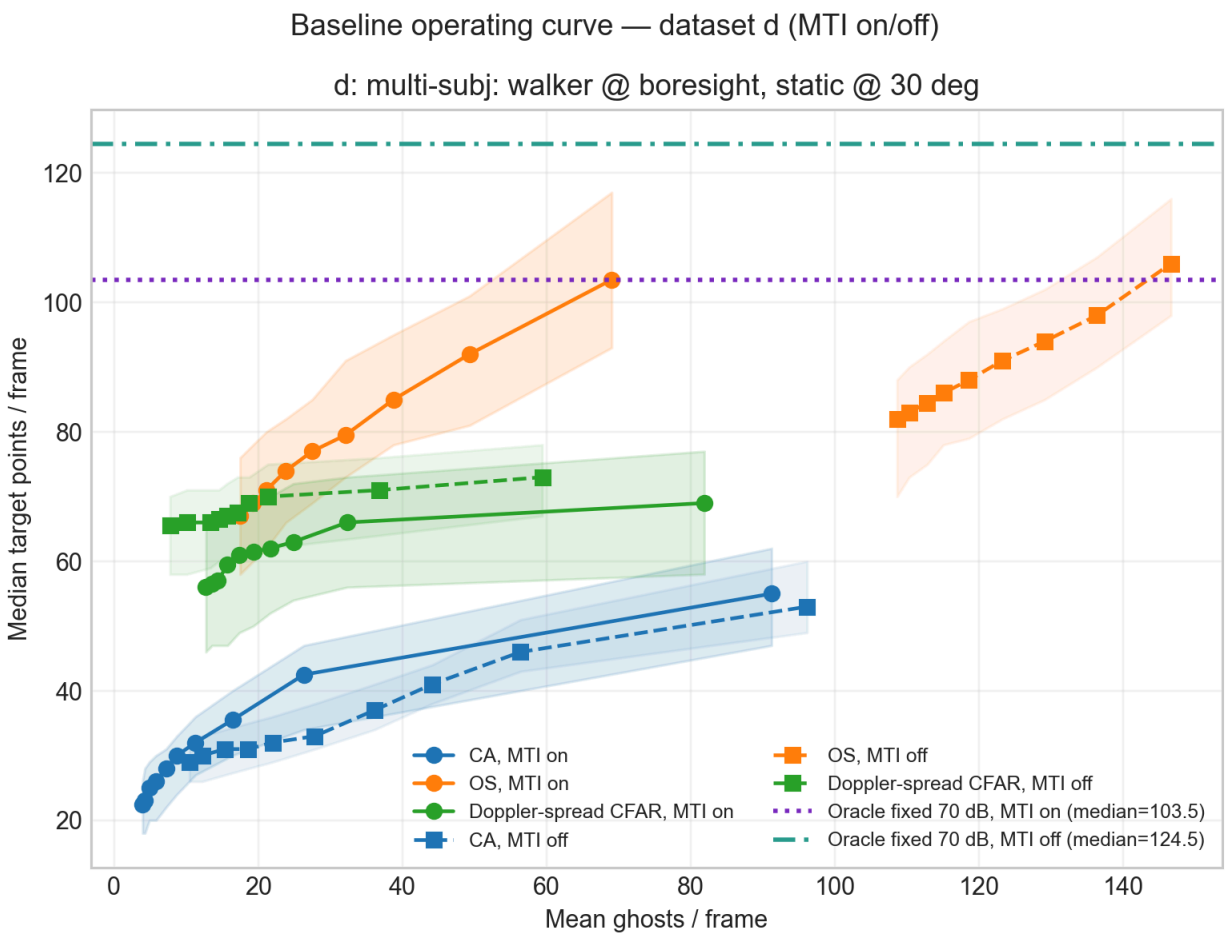


Fig. 4.2: dataset d operating curve

8.4 Static scenes observation

The static single-subject scene (dataset g) produces a differently shaped curve from the dynamic scenes. As expected from the simplicity of scenes, points are gained almost immediately at close to zero ghost points, with OS-CFAR and DS-CFAR performing comparably and CA-CFAR clearly worse.

This is consistent with the rest of the picture: with no moving extended target spreading energy across the Doppler axis, the reference window contamination that drives the loss in the dynamic scenes largely disappears, and the choice of detector matters much less.

8.5 Sensitivity to static clutter removal

Turning MTI off significantly increases the ghost count for OS-CFAR. The mechanism is direct: with static clutter left in the map, the clutter returns are themselves cells under test carrying enough power to cross the threshold, so OS-CFAR detects them. Those detections fall outside the person depth cloud and are scored as ghosts.

Another issue is MTI when it is switched on, quasi-static subject loses SNR and drops below the local threshold and quasi static targets go undetected due to target masking. The two costs sit on opposite

sides of the MTI decision, which is why both states are reported throughout this report rather than fixing one. The figures below show this effect, as well as a good visual comparison of DS CFAR performing and detecting better than OS CFAR.

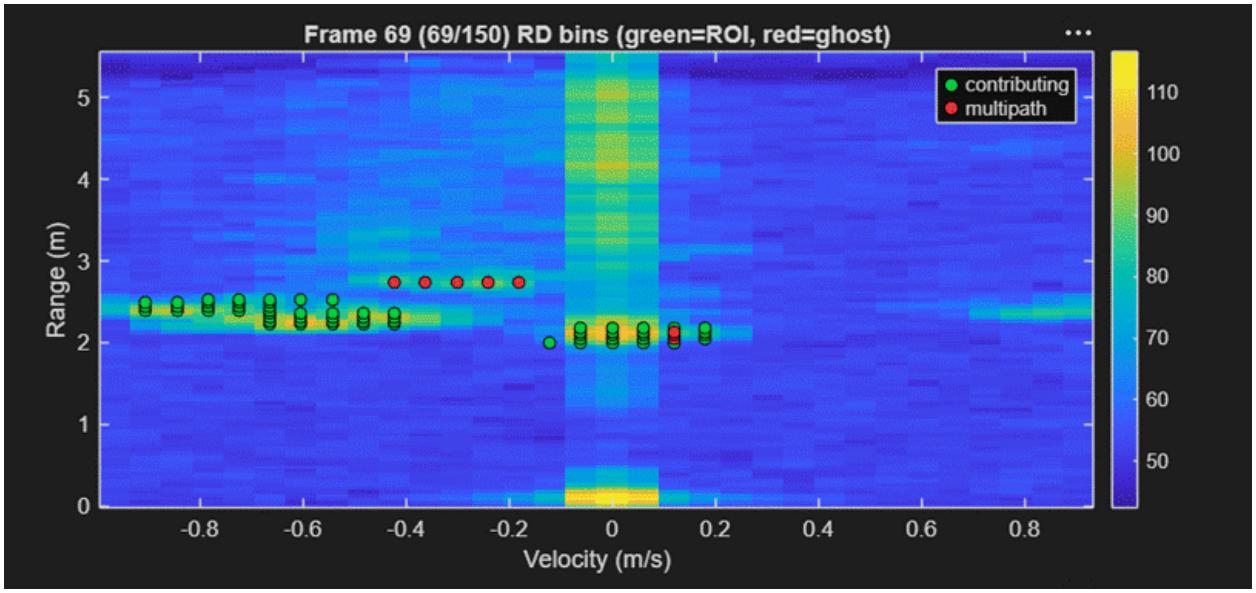


Fig 5.1: Doppler Spread CFAR Static removal off. Higher SNR at 0 doppler, hence more detections

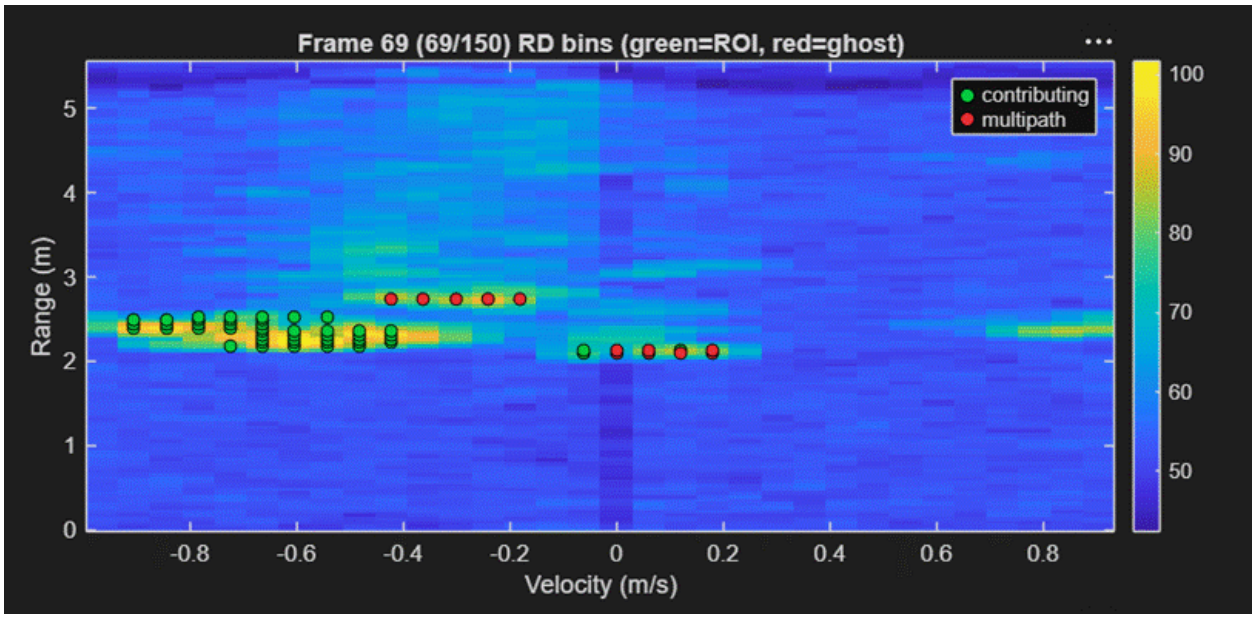


Fig 5.2: Doppler Spread CFAR Static removal on. Less SNR at 0 doppler due to MTI; mis detected and less detected points.

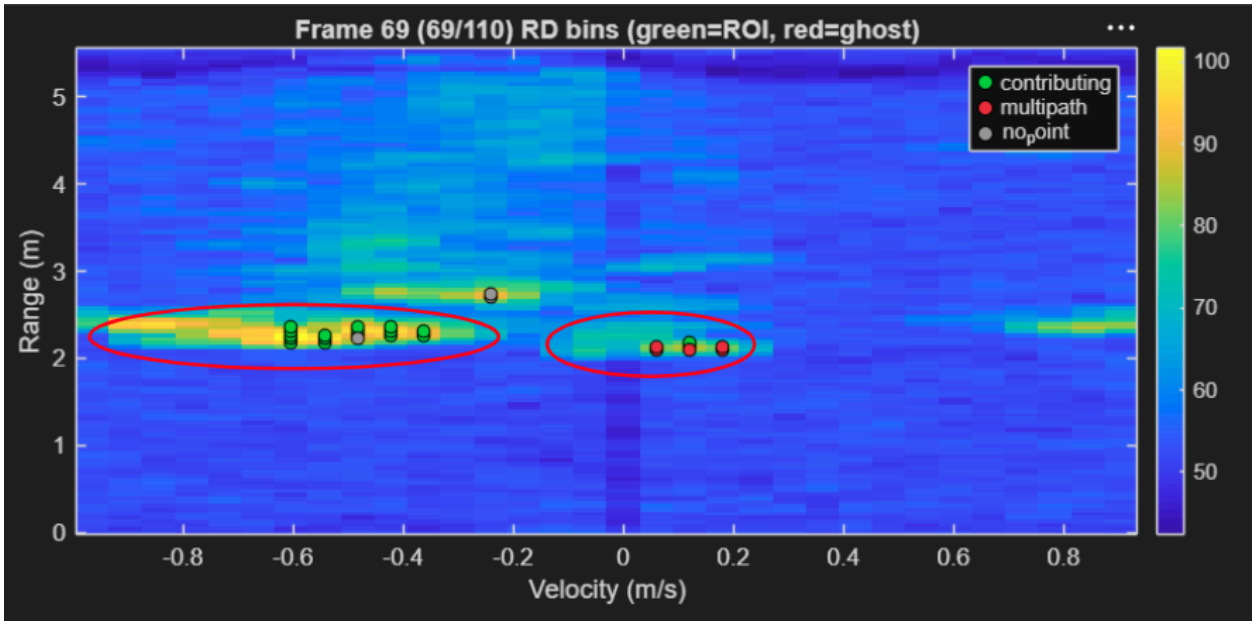


Fig 5.3: OS CFAR Static removal on. Less SNR at 0 doppler due to MTI; mis detected and less detected points.

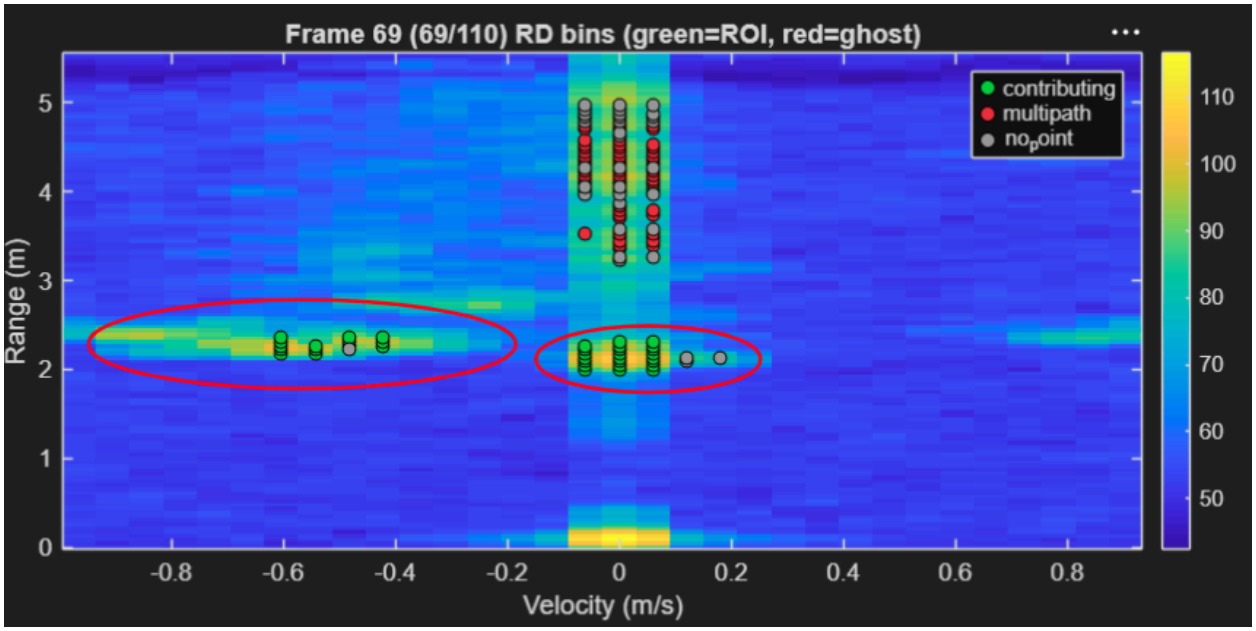


Fig 5.4: OS CFAR Static removal off. Higher SNR at 0 doppler, hence more detections

A secondary observation, offered as such rather than as a result this study set out to establish: DS-CFAR is close to invariant to the MTI state, with similar operating curves for MTI on and MTI off. A plausible reading is that Doppler condensation helps on both sides of the tradeoff — integrating a weakened quasi-static return rather than thresholding it bin by bin, while not responding to static clutter the way a

per-cell 2D detector does: because the range-only detector collapses the Doppler axis, and static clutter sits at comparable energy across ranges, it does not present the isolated high-power cells that a per-cell detector would flag. This was not anticipated and was not what the experiment was looking for.

It should not be read as robustness to static clutter. The configuration that should break it is a target occupying the same range bin as strong static clutter, since a range-only detector has no mechanism to separate the two once the Doppler axis is collapsed. None of the six datasets contain that case, so it is untested. The observation is recorded because it points somewhere worth following, not because it is settled.

9. What this work does not measure but may need to be evaluated

Failure due to exponential clutter PDF violation from Section 2 is stated in this report but not quantified. The nominal PFA is a design parameter derived from the closed-form threshold, which assumes exponentially distributed clutter. Indoor multipath and multitarget returns violate that assumption in the direction of heavier tails, and this violation acts on the operating point through two mechanisms that push realized PFA in opposite directions. From the cell-under-test perspective, a clutter CUT drawn from a heavy-tailed PDF exceeds an exponential-designed threshold more often than the design predicts, which raises realized PFA; this is the classical distribution-mismatch behaviour analysed for OS-CFAR under Weibull and other two-parameter backgrounds [12, 13], where the scale factor must be corrected for the shape parameter or the maintained-PFA property is lost. From the reference-cell perspective, a heavier tail raises the probability that a reference cell holds a high-power sample, which inflates the local estimate Z and raises the threshold $T = \alpha \cdot Z$; this primarily lowers detection probability (the same masking mechanism as the Doppler skirt and the quasi-static MTI loss) and, as a side effect, lowers realized PFA. For OS-CFAR the reference-cell effect is attenuated: with an order-statistic estimator at rank fraction 0.75, up to roughly a quarter of the reference cells can be high-power spikes before the threshold statistic itself is contaminated, so the CUT-side (PFA-raising) effect is comparatively less opposed than it would be under cell averaging. Because the net direction of realized PFA is therefore scene-dependent and not fixed by the nominal value, every comparison in Section 8 is made against realized mean ghost counts rather than against nominal PFA — the nominal value is used only to sweep the operating point, not as a claim about false alarm probability.

10. Conclusions and direction

1. The information loss at the CFAR stage is real and now measured rather than asserted. A substantial gap exists between conventional CFAR output and the recoverable reference level across all dynamic and mixed scenes.
2. The loss is recoverable but expensive. OS-CFAR reaches the oracle target count only at a false alarm load that makes the point cloud unusable downstream. The problem is a cost curve, not a hard limit.
3. A potential problem may be a fixed two-dimensional window geometry applied to a target whose Doppler extent is not predictable.

4. Decoupling range from Doppler is the right structural response given the human scattering profile, and the fixed integration window is what limits the existing instance of it. The Doppler-width sweep is the direct evidence: widening the window recovers the target points the baseline configuration left behind.
5. A secondary and unanticipated observation, not an established result: DS-CFAR is close to invariant to the static clutter removal state. It is untested against the configuration that should break it, and is recorded as a direction rather than a finding.

The directions that follow:

- **Adaptive, target-aware Doppler condensation.** Replace the fixed window with something that tracks the actual Doppler occupancy of the target. This is where the critique in Section 8.3 becomes a method rather than a complaint.
- **Heavy-tailed clutter branch.** Decide whether the indoor problem is better attacked by fixing the statistic geometry (dimension 1) or the distributional assumption (dimension 2), or whether the two compose. This is also the route to quantifying failure mode due to violation of the exponential PDF assumption.
- **Static clutter stress case.** Collect a scene where a quasi-static subject shares a range bin with strong static clutter, which is the configuration that should break the DS-CFAR MTI invariance reported in Section 8.5.
- **Downstream validation.** Whether recovered skirt points measurably improve HAR or mesh reconstruction accuracy is the motivating hypothesis for this entire line of work and remains untested. The experiment is a fixed downstream model fed point clouds from each detector configuration.

11. Limitations

- Synchronization was software-based, not hardware-triggered; dataset-specific time offsets were set manually by inspection.
- The oracle is a diagnostic upper reference, not a practical detector. Its 70 dB value is chosen by inspection of the data rather than derived, so the reference level it marks is an estimate of recoverable target energy and not a proven bound.
- The evaluation is six short recordings from a limited set of subjects and trajectories.
- The shared AoA and DBSCAN stage makes relative rankings fair but leaves AoA noise in the absolute point counts.

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